

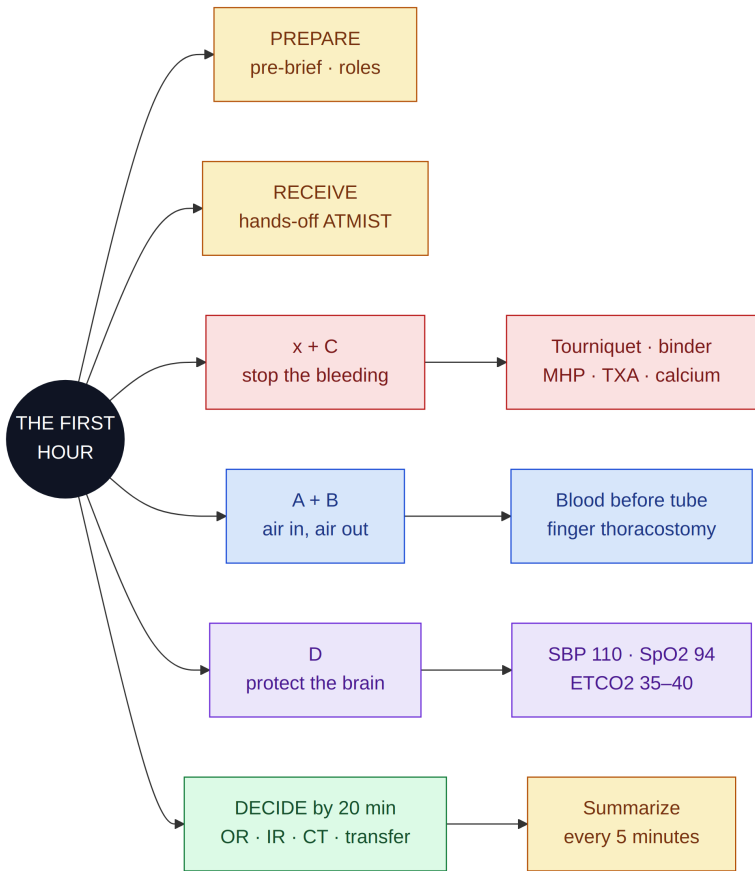
Trauma Resuscitation: The First Hour — An Alberta Team Leader's Playbook

Sep 23, 2026 · Dr. AJ Esterhuizen

The first hour at a glance

Every section opens with a **brain map** and closes with a **study box**. The colours are the same everywhere, so your eye learns where to look.

Colour	Means
Red	Bleeding and circulation
Blue	Airway and breathing
Purple	Brain and spine
Green	Decisions and destinations
Amber	Team and communication



How to study this: read the map first, then the section, then cover the section and redraw the map from memory. Say the scripts out loud in the room where you work — context-dependent recall is strongest where you'll need it.

1. How to use this playbook

The team leader's first hour in trauma comes down to one question asked over and over: **is this patient bleeding (or herniating, or suffocating) to death, where, and what is the fastest way to stop it?** Everything below — scripts, checklists, dose tables — exists to answer that question out loud, in order, on time.

Medical resuscitation feels scripted because ACLS gives you a loop: rhythm, shock, drug, 2 minutes, repeat. Trauma feels unscripted because the work is **horizontal** (six things happen at once), the diagnosis is **anatomic** (where is the hole?), and **the ED is a waystation** (the definitive fix is usually in the OR, IR suite, or another hospital). The fix is to give yourself a loop anyway.

The team leader's loop



The summarize step is the one most people skip, and it is the one that makes you sound in control. It is your ACLS "2-minute check": a forced pause to state what you know, what you've done, and what happens next.

Core mental models

- **xABCDE, not ABC.** Exsanguinating external hemorrhage is managed before the airway. ATLS 11th edition adopted this sequence (see Section 2). The military equivalent is MARCH (Massive hemorrhage, Airway, Respiration, Circulation, Head/Hypothermia).
- **Circulation before intubation in shock.** "Resuscitate before you intubate" — a hypotensive trauma patient often arrests on induction because positive pressure kills preload. Blood first, tube second, unless the airway itself is the problem.
- **Hands off, eyes up.** The leader does not do procedures unless there is nobody else. The moment you pick up a scalpel, nobody is running the room. If you must do a procedure, say out loud who is taking over leadership.
- **Hypotension in trauma is hemorrhage until proven otherwise.** Tension pneumothorax and tamponade are the two obstructive exceptions you actively exclude. Neurogenic shock is a diagnosis of exclusion.
- **Five places to lose blood: "on the floor and four more"** — external, chest, abdomen, pelvis/retroperitoneum, long bones (femur).
- **The lethal diamond:** hypothermia, acidosis, coagulopathy, hypocalcemia. Every intervention either feeds it or starves it.
- **Time is the currency.** Median time to death from hemorrhage is measured in hours but most preventable deaths happen in the first one. Your ED dwell time for an unstable patient should be minutes, not an hour.

How the document is organized

Sections 3–5 are chronological: before arrival, handover, primary survey, with the words to say at each step. Sections 6–11 are deep dives (hemorrhage, airway, chest, arrest, head, special populations). Section 12 covers the decision points that end the first hour. Section 14 is a phrase bank for the team leader, and Section 15 is a one-page card. Doses are adult unless marked; always cross-check against your local AHS protocols and pharmacy.

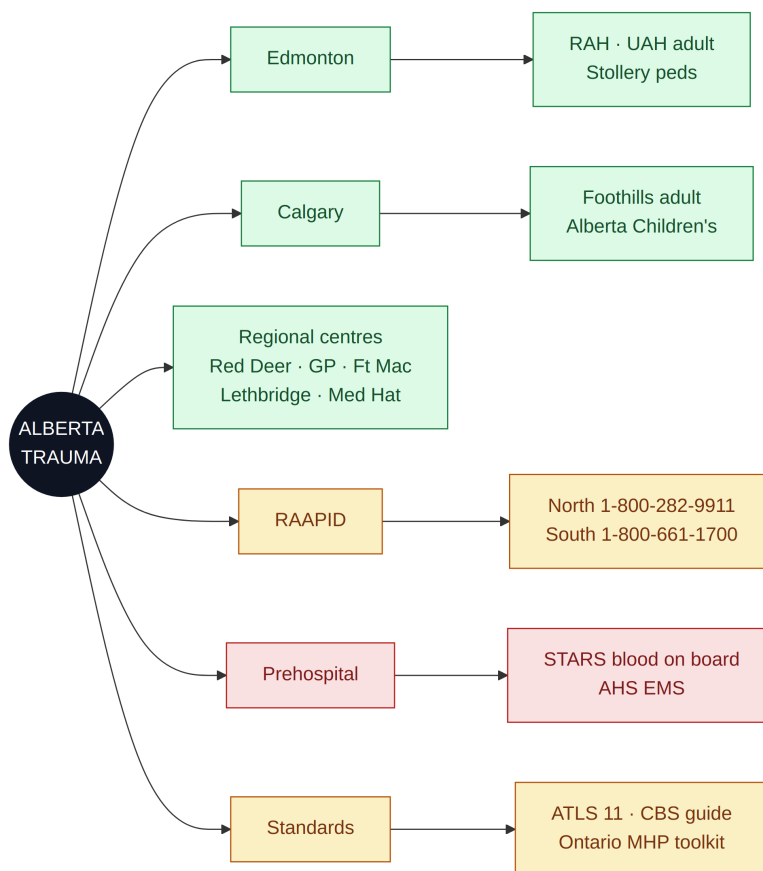
Memory hook — "On the floor and four more": external, chest, abdomen, pelvis, long bones.

Pearl — If you can't say where the patient is going next, the resuscitation isn't finished.

Pitfall — Picking up a procedure means putting down leadership. Hand it over out loud.

2. The Alberta and Canadian context

The RAH is one of two adult lead trauma centres in Edmonton (with the UAH), inside a 10-centre provincial network now run by **Trauma Care Alberta**, a provincial program under Acute Care Alberta ([Trauma Care Alberta](#)). Knowing the system tells you who to call and where the patient goes next.



The Alberta trauma network

Region	Trauma centres
Edmonton & area	Royal Alexandra Hospital, University of Alberta Hospital (adult); Stollery Children's Hospital (pediatric)
Calgary & area	Foothills Medical Centre (adult); Alberta Children's Hospital (pediatric)
Central	Red Deer Regional Hospital
Northern	Grande Prairie Regional Hospital; Northern Lights Regional Health Centre (Fort McMurray)
Southern	Chinook Regional Hospital (Lethbridge); Medicine Hat Regional Hospital

Source: [Trauma Care Alberta service locations](#).

- **Designation status:** The Edmonton and Calgary adult and pediatric centres were designated Level I or II under the Trauma Association of Canada (TAC) standard. There has been no national verification process for several years; TAC is moving toward the American College of Surgeons Committee on Trauma (ACS-COT) standard ([CBC, Jan 2025](#)).
- **RAH Trauma Medical Service** runs 24/7/365 ([AHS: RAH Alberta Trauma Services](#)). The RAH trauma program is led by Dr. Matt Strickland (per CBC, Jan 2025).
- **Pediatrics:** In the Edmonton zone, major trauma aged 16 and under goes to the Stollery; 17 and over to RAH or UAH ([Edmonton 2010 Trauma Report](#)). Confirm current age cut-offs locally.

- **Alberta Trauma Registry** captures patients with ISS ≥ 12 (or penetrating trauma needing OR resources) who are admitted or die in the ED after treatment starts. Your documentation feeds it.

Transfers and advice: RAAPID

RAAPID is the single number for inter-facility trauma referral, consults and destination decisions, partnered with STARS and AHS EMS ([AHS RAAPID](#)).

Line	Covers	Numbers
RAAPID North	Patients north of Red Deer	1-800-282-9911 · 780-735-0811
RAAPID South	Red Deer and south	1-800-661-1700 · 403-944-4486

Prehospital blood in Alberta

STARS was the first air medical service in Canada to carry blood; Blood on Board now runs at all six bases ([STARS history](#)). STARS later added a prehospital massive hemorrhage protocol with additional hemostatic medications ([STARS, 2023](#)). Since February 2025, the Winnipeg base carries two units of whole blood as part of the TOWAR trial ([STARS, 2025](#)). **Practical point:** at handover, always ask "what blood and TXA did they get, and when?" so your MHP count and TXA timing start from the right number.

The guidelines this playbook draws on

Resource	What it gives you
ATLS 11th edition (ACS, 2025)	xABCDE sequence, 1:1:1 balanced resuscitation, context-based airway, new team communication and trauma systems chapters
Canadian Blood Services Clinical Guide, Ch. 11	Canadian MHP principles, TXA, fibrinogen, emergency blood selection
Callum et al., CMAJ Open 2019	Canadian Delphi consensus on MHP principles and quality indicators
Treat the Bleed	Canadian bleeding-management decision support
CBS LrWB FAQ	Whole blood availability in Canada (trials and military only)
Canadian CT Head Rule; Canadian C-Spine Rule (Stiell et al.)	Validated Canadian imaging decision rules (Section 13)
Brain Trauma Foundation 4th ed.; EAST guidelines	TBI targets; thoracotomy and damage control guidance
TRIK course (Royal College)	Canadian pediatric trauma resuscitation course

ATLS 11 also introduced a new trauma systems chapter and expanded team communication content ([ACS, Sept 2025](#)). If your last ATLS was the 10th edition, the xABCDE change is the headline difference.

Memory hook — North of Red Deer? **282-9911**. Red Deer and south? **661-1700**.

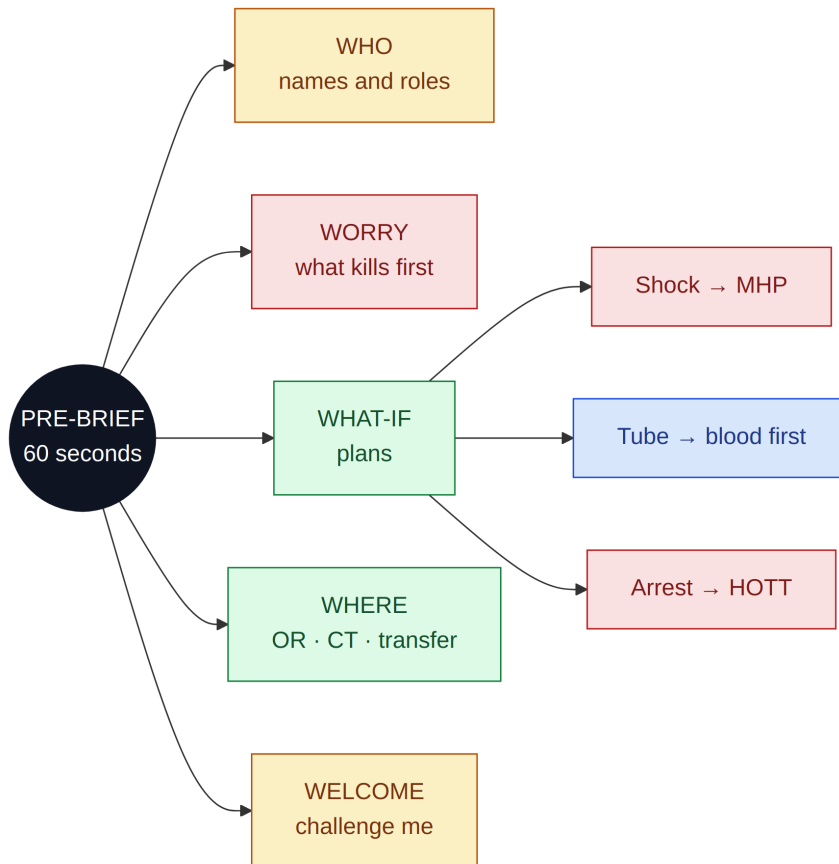
Pearl — Ask every crew: "What blood and TXA, and at what time?" Your MHP count and the 3-hour TXA window start in the field.

Pitfall — ATLS 10 habits. ATLS 11 moved exsanguinating hemorrhage (x) ahead of the airway.

3. Before the patient arrives (minus 10 to 0)

The best trauma resuscitations are won before the doors open. A 60-second pre-brief, named roles and a pre-agreed "if-then" plan remove most of the chaos you feel later.

The pre-brief in four Ws — plus one invitation:



Step 1 — Get the pre-alert and build a picture

Write on the whiteboard: age, sex, mechanism, time of injury, vitals trend, interventions (blood, TXA, needle decompression, airway), ETA. Start the clock from **time of injury**, not arrival — TXA and the "golden hour" both run from injury.

Ask yourself three questions:

1. **What will kill this patient first?** (bleeding, airway, head, chest)
2. **Where is the patient going after me?** (OR, IR, CT, ICU, UAH for a specialty, Stollery)
3. **Who do I need in the room or on the phone right now?** (trauma surgery, anesthesia, neurosurgery, OB, blood bank)

Step 2 — The pre-brief script (say it out loud)

"Can I have everyone's attention for sixty seconds — eyes up please. I'm AJ, I'm team leader today.

We have a [age] [sex], [mechanism], injured at [time]. EMS reports [vitals]. They've had [blood / TXA / interventions]. ETA [x] minutes.

My main worry is [hemorrhage / head injury / airway / chest].

Roles: [Name], airway. [Name], primary survey and procedures. [Name], left side — access and blood. [Name], right side — monitors and meds. [Name], you're recording and calling out times. [Name], rapid infuser. [Name], you're our blood runner.

The plan: if they arrive hypotensive, I'll call MHP immediately — blood through the rapid infuser, TXA if not already given, calcium with the first units. If they need a tube, we resuscitate first, then ketamine [dose] and rocuronium [dose], video laryngoscope, bougie, cric kit open. If they arrest, we go to the traumatic arrest pathway — bilateral finger thoracostomies, blood, eFAST for tamponade.

Trauma surgery is [aware / in the room]. Please speak up if you see something I've missed — I want to hear it. Any questions or concerns?"

Then: name stickers on chests, gloves and lead on, everyone at their station.

Step 3 — Assign roles

Role	Position	Core jobs
Team leader	Foot of bed, hands off	Directs, summarizes, decides, calls consultants
Airway physician/RT	Head	C-spine, airway, ventilation, ETCO2
Primary survey / procedures	Patient's right	Hands-on xABCDE, eFAST, chest decompression, lines
Nurse 1 (access & blood)	Patient's left	Large-bore IV/IO, bloods, rapid infuser, product checks
Nurse 2 (monitoring & meds)	Patient's right	Monitors, meds, TXA, calcium, temperature
Recorder	Wall/whiteboard	Times, vitals every 5 min, products given, time since TXA
Blood runner	Moving	Brings MHP packs, returns empty cooler
Trauma surgery	Joins team leader or procedures	Operative decisions, definitive care
Social work / family liaison	Outside room	Family, belongings, police

In a thin-staffed night at RAH or NECHC you may combine roles. Say out loud who is doing two things.

Step 4 — Room and equipment checklist



Room warm (turn thermostat up), forced-air warmer on bed, warm blankets



Rapid infuser primed; blood warmer on; pressure bags as backup

- ☐ Uncrossmatched O blood requested to the bedside if shock is likely (O-negative for females of childbearing potential and children)
- ☐ Tranexamic acid 1 g drawn up; calcium chloride 1 g drawn up
- ☐ Pelvic binder at bedside, open; commercial tourniquets x2
- ☐ Airway: video laryngoscope, direct laryngoscope, bougie, two suctions, ETCO2, SGA, surgical cric kit (scalpel-finger-bougie)
- ☐ RSI drugs drawn: ketamine, rocuronium, push-dose vasopressor, post-intubation analgesia/sedation
- ☐ Chest: scalpel, curved forceps, chest tube tray, 28–32 Fr tubes (smaller is acceptable for isolated pneumothorax)
- ☐ Thoracotomy tray located and ready if penetrating trauma or arrest is possible
- ☐ Ultrasound on, probe selected, gel out
- ☐ Trauma labels/ID band ready, blood bank sample tubes pre-labelled
- ☐ PPE for everyone; lead aprons if plain films are planned
- ☐ Portable X-ray called; CT informed if the patient may go straight there
- ☐ Crowd control: tape line or designated "stand here" zone for observers

Step 5 — Pre-activate what you can

- **Blood bank:** "We have a possible MHP arriving in ten minutes, please have the first pack ready."
- **Trauma surgery:** Page early. Late is worse than unnecessary.
- **CT/IR/OR:** For a patient likely to need surgery, a heads-up to the OR charge nurse saves minutes.

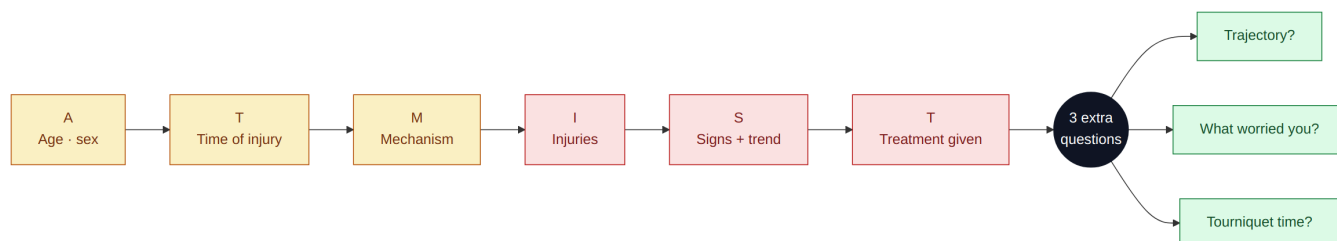
Memory hook — Who · Worry · What-if · Where · Welcome.

Pearl — Stand at the foot of the bed before the patient arrives. Where you stand at minute zero is where the team will look for you.

Pitfall — Skipping the pre-brief because "there isn't time." It takes 60 seconds and saves ten minutes of confusion.

4. Minute 0-2: Receiving the patient

A structured, hands-off handover takes 30-60 seconds and prevents the single biggest information loss in trauma. Protect it — unless the patient is actively dying.



Decide: handover first, or move first?

Patient state	Sequence
Talking, SBP > 90, not peri-arrest	Hands-off handover on the EMS stretcher, then transfer
Shocked, deteriorating, or airway at risk	Transfer immediately; EMS gives handover while the team works; only procedures that cannot wait
Cardiac arrest	Transfer with CPR; EMS gives "ATMIST" in 15 seconds; go straight to traumatic arrest pathway (Section 9)

The words

"Everyone, hands off for thirty seconds please — paramedics, go ahead."

Then let them talk uninterrupted. Hold questions until they finish.

ATMIST (the structure to expect, and to prompt for if missing):

Letter	Content	Prompt if missing
A	Age, sex	—
T	Time of injury	"What time did this happen?"
M	Mechanism	"Speed, restraint, extrication time, height of fall, weapon?"
I	Injuries found or suspected	"Anything on your primary survey?"
S	Signs: vitals and trend, GCS, pupils	"First and worst BP? GCS at scene?"
T	Treatment given	"Blood, TXA, needle or finger decompression, airway, meds, tourniquet times?"

Then ask the three questions paramedics are often not asked:

1. "What were they like at the scene compared with now?" (trajectory matters more than the snapshot)
2. "What worried you most?"
3. "Any tourniquet on, and what time?"

The transfer

"Airway has control of the head. On my count — ready, steady, slide."

The airway person controls the head and calls the move in most departments. Use a slide sheet. Log-roll is for the back exam later, not the transfer.

Immediately after the transfer (first 60 seconds, in parallel)

- Monitors on: SpO2, 3-lead, BP cycling every 2-3 min, ETCO2 if intubated
- High-flow oxygen
- Two large-bore IVs or IO; bloods including group and screen and blood gas
- Expose while talking; keep warm
- Recorder starts the clock on the whiteboard

"Thank you. Airway — tell me about the airway."

This last line is your transition into the primary survey. It tells the team you are now running the resuscitation.

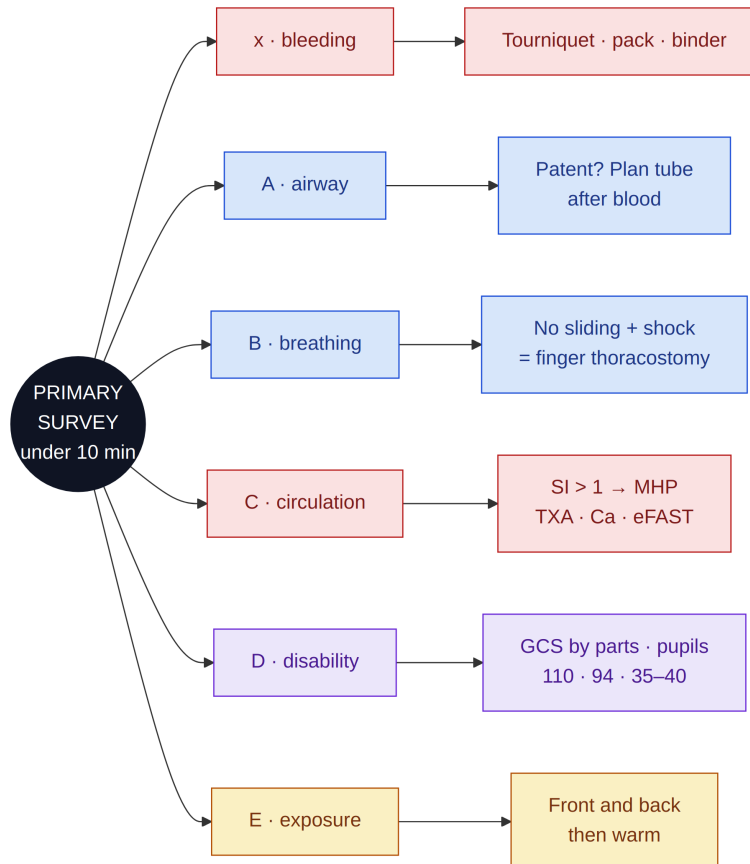
Memory hook — Hands off, eyes up, mouths shut: 30 seconds buys the whole story.

Pearl — Trajectory beats snapshot. "SBP 110 at scene, 85 now" is a different patient from "85 at scene, 110 now."

Pitfall — Asking questions mid-handover. Crews lose their place and drop the treatment list — the part you most need.

5. Minutes 2-10: The primary survey, with the words

The primary survey is sequential in your head and simultaneous in the room. Your job is to pull each finding out loud, attach an action to it, and close the loop. The primary survey should be done in under 10 minutes; if it isn't, something is being missed or someone is doing a procedure that should wait.



The pattern for every letter is the same: **ask → hear the finding → state the action → confirm it's done.**

"Airway, what have you got?" → "Talking, patent." → "Great, airway's patent. Keep the collar on. Breathing?"

x — Exsanguinating external hemorrhage

Look for: pooling blood, soaked dressings, amputations, groin/axilla/neck wounds, scalp lacerations (they bleed far more than expected).

Do:

- Limb: tourniquet high and tight, tighten until bleeding stops and distal pulse is gone; second tourniquet above the first if needed. Record the time on the tourniquet and the whiteboard.
- Junctional (groin, axilla, neck): direct pressure with a finger in the wound, then wound packing with hemostatic gauze and 3 minutes of firm pressure. Foley balloon tamponade for deep penetrating tracts.
- Scalp: staples or running suture now if it is pouring — it can cause shock by itself.
- Pelvis: binder over the greater trochanters (not the iliac crests) for any shocked patient with a mechanism for pelvic injury. Do not spring the pelvis.

Say:

"Any major external bleeding?" ... "Tourniquet on the left thigh, applied at 14:32 — recorder, please note the time."

A — Airway with cervical spine protection

Look for: talking in full sentences (airway patent, GCS likely > 8), stridor, gurgling, facial fractures, neck hematoma, burns to face, blood in the mouth.

Do: suction, jaw thrust, adjuncts. Manual in-line stabilization for any intervention. Decide early: does this patient need a definitive airway, and **when**?

Say:

"Airway, what have you got?"

If patent: "Airway patent. Maintain in-line stabilization."

If threatened: "This airway's at risk. We're going to intubate — but I want two units in first unless it deteriorates. Airway, prepare for RSI; cric kit open."

Indications for a definitive airway: failure to maintain or protect (GCS $\leq$ 8 is a guide, not a rule), expected deterioration (inhalation burns, expanding neck hematoma, facial fractures with bleeding), failure of oxygenation or ventilation, combative patient needing CT who can't be safely sedated otherwise. See Section 7.

B — Breathing and ventilation

Look for: RR, SpO₂, chest rise symmetry, paradoxical movement (flail), open wounds, crepitus, tracheal deviation (late), distended neck veins (absent if hypovolemic), breath sounds, percussion.

Ultrasound: lung sliding bilaterally (absence = pneumothorax until proven otherwise), hemothorax on the RUQ/LUQ views above the diaphragm.

Do: oxygen, decompress tension physiology immediately (finger thoracostomy 4th/5th ICS anterior to mid-axillary line), seal open chest wounds with a vented seal, chest tube for hemothorax.

Say:

"Breathing — what are you finding?" ... "Absent sliding left, SpO₂ 88, hypotensive — this is a tension until proven otherwise. [Name], finger thoracostomy left side now please. Tell me when it's done and what you feel."

C — Circulation with hemorrhage control

Look for: HR, BP, shock index (HR/SBP > 1 is concerning, > 1.4 very concerning), mental status, skin, capillary refill, central vs peripheral pulses. Remember that young patients, athletes, beta-blocked and elderly patients hide shock. Lactate and base deficit are your early warning.

Find the bleeding — "on the floor and four more":

Space	How you find it in the resus bay
Floor (external)	Look, logroll later, check the stretcher
Chest	eFAST, chest X-ray, chest tube output
Abdomen	eFAST (RUQ, LUQ, pelvis), exam; CT if stable
Pelvis / retroperitoneum	Pelvic X-ray, mechanism, binder; CT or angio
Long bones	Femur deformity (up to 1.5 L per femur); traction splint

Do: two large-bore IVs (or IO — humeral head is excellent for flow), bloods, activate MHP when criteria are met (Section 6), TXA if within 3 hours, calcium, warm everything.

Say:

"Circulation — HR 130, BP 85 over 50, shock index 1.5. He's in hemorrhagic shock. I'm activating the massive hemorrhage protocol. [Name], please call blood bank now. [Name], give TXA 1 gram. First two units through the rapid infuser, calcium with them. eFAST now please."

eFAST callout:

"Talk me through the FAST — RUQ ... LUQ ... pelvis ... pericardium ... lung sliding left and right."

A positive FAST in an unstable patient is an OR decision, not a CT decision. A negative FAST does not exclude bleeding (retroperitoneum, pelvis, early small bleeds).

D — Disability

Look for: GCS (report as E, V, M components — motor score is the most useful), pupils (size, symmetry, reactivity), lateralizing signs, glucose, spinal cord signs (priapism, sensory level, flaccid limbs with hypotension and bradycardia).

Do: avoid hypoxia and hypotension (the two things that double TBI mortality), check glucose, early CT head, treat herniation signs (Section 10).

Say:

"Disability — GCS?" ... "E1 V2 M4 — GCS 7, left pupil 5 millimetres and sluggish. This is a significant head injury with possible herniation. Targets: SBP above 110, sats above 94, ETCO2 35 to 40. Can someone call neurosurgery? Let's get hypertonic saline ready."

E — Exposure and environment

Do: cut all clothes off, look at the whole patient (front, and back during the log-roll), then cover and warm. Temperature measurement. Remove jewellery on injured limbs. Check for a medic alert, pregnancy, anticoagulants.

Say:

"Exposure — let's get everything off, check front and back, then get him warm. Temperature please. Room temperature up."

Log-roll timing: do it once, as part of the secondary survey, with the team leader directing. Unnecessary log-rolls in pelvic fractures dislodge clot.

The primary survey adjuncts (order at the end of C, run in parallel)

Adjunct	Why
Blood gas with lactate, base deficit, Hb, K+, ionized calcium	Shock severity, hypocalcemia, early trend
CBC, INR, PTT, fibrinogen, lytes, creatinine, group and screen, β -hCG, ethanol	Baseline for MHP targets
Chest X-ray (supine AP)	Hemothorax, pneumothorax, mediastinum, tube positions
Pelvic X-ray	Unstable pelvis in the shocked patient going to OR/IR (skip if going straight to CT and stable)
eFAST	Free fluid, pneumothorax, tamponade
ECG	Blunt cardiac injury, pre-existing disease
Foley (unless urethral injury suspected) and gastric tube	Output monitoring, decompression

The end-of-primary-survey summary (say this every time)

This is the most important thing you will say. It forces decisions and resets the room.

"Can I have a moment, everyone. Here's where we are:

*This is a [age] [mechanism] at [time since injury]. **A** is [patent / secured]. **B**: [findings, interventions]. **C**: [vitals, shock class, FAST result, products given]. **D**: [GCS, pupils]. **E**: [findings, temperature].*

My leading concern is [bleeding from the abdomen / tension / head injury]. The plan is [OR / CT / IR / more resuscitation]. Next steps: [three concrete tasks with names].

Does anyone have anything to add, or anything I've missed?"

Then repeat a shorter version every 5 minutes or with every change (Section 12).

Memory hook — Ask → hear → act → confirm. Every letter, every time.

Pearl — Unstable + positive FAST = OR, not CT.

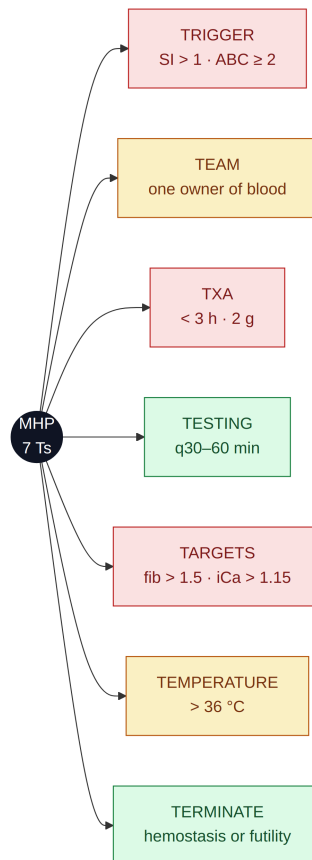
Pitfall — A negative FAST does not exclude bleeding — pelvis and retroperitoneum hide from ultrasound.

Pitfall — Any deterioration means back to the top: "Stop. Airway?"

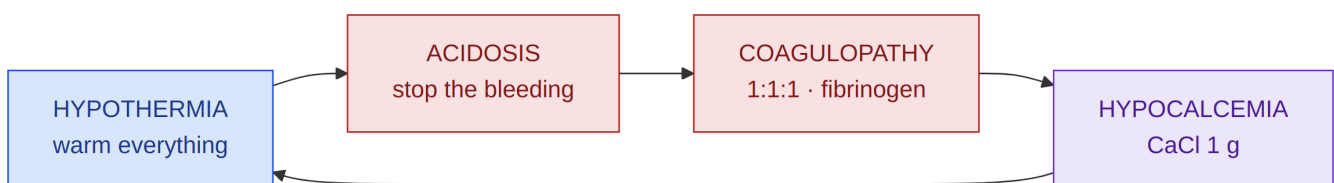
6. Hemorrhage control and the massive hemorrhage protocol

In the bleeding trauma patient, the ED's job is to buy time with balanced blood, TXA, calcium and warmth while moving the patient to definitive hemorrhage control. Every one-minute delay to the first MHP cooler is associated with about a 5% increase in mortality ([CBS Clinical Guide, Ch. 11](#)).

The 7 Ts wheel:



The lethal diamond — each corner feeds the next:



Damage control resuscitation in four parts

1. **Stop the bleeding early** — damage control surgery, IR, pelvic binder, tourniquets.
2. **Avoid crystalloid** — ATLS limits it to about 1 L before switching to blood; large volumes worsen coagulopathy.
3. **Limit the pressure** — permissive hypotension until hemorrhage control, *except* in brain or spinal cord injury.
4. **Give balanced blood** — plasma, platelets and red cells approximating whole blood (1:1:1 or near it). PROPPR showed no overall 24-h or 30-day survival difference for 1:1:1 vs 1:1:2, but less death from exsanguination at 24 hours.

The 7 Ts of an MHP (Canadian framework)

The Ontario provincial MHP toolkit organizes the protocol into seven Ts: triggering, team, tranexamic acid, testing, transfuse to target, temperature and termination ([Ontario MHP survey, CJEM 2025](#)). Use it as your mental checklist.

T	What to do	What to say
Trigger	Activate on clinical criteria; don't wait for labs	"I'm activating the massive hemorrhage protocol."
Team	One person owns the blood: calls blood bank, tracks products	"[Name], you own the blood. Tell me every 4 units."
TXA	Within 3 h of injury, ideally within 1 h	"TXA 2 grams IV now, if EMS hasn't given it — confirm?"
Testing	Baseline then roughly every 30–60 min	"Send MHP bloods now and every 30 minutes."
Transfuse to target	Ratio first, then lab-guided	"Fibrinogen is 1.2 — give 4 grams fibrinogen concentrate."
Temperature	Measure within 15 min; everything warmed	"What's his temperature? Warm the room."
Termination	Hemostasis achieved, or futility	"Bleeding is controlled; I'm stopping the MHP — please call blood bank."

When to trigger

Clinical judgement alone is only about 65% sensitive and specific. Use objective aids ([CBS Clinical Guide, Ch. 11](#)):

- **Shock index** > 1 (HR/SBP), especially after 1 L of fluid or blood
- **ABC score** ≥ 2: penetrating mechanism, SBP ≤ 90, HR ≥ 120, positive FAST (one point each)
- **Critical administration threshold**: ≥ 3 units of RBC in 1 hour
- **Gestalt red flags**: amputation, pelvic fracture with shock, positive FAST with hypotension, chest tube > 1,500 mL initial or > 200 mL/h

If in doubt, activate. You can always stop it.

Emergency blood selection

- **Group unknown**: uncrossmatched group O red cells. O-negative for females of childbearing potential and children; O-positive acceptable for males and older females. Ontario MHP 2.0 defines childbearing potential as age < 45 ([ORBCoN MHP 2.0](#)); **check the AHS age cut-off**.
- **Plasma**: AB (or group A in some protocols) until group known.
- **Switch to group-specific** as soon as the group is confirmed on a second sample — regardless of how many O units have gone in.
- **Draw the group and screen before the first unit** if at all possible, and label it perfectly. ABO errors are more common in chaotic trauma.
- **Whole blood (LrWB)**: Canadian Blood Services has produced it since November 2022, but it is available only to the Canadian Armed Forces and to hospitals in clinical trials ([CBS LrWB FAQ](#)).

Tranexamic acid

Regimen	Details
CRASH-2	1 g IV over 10 min, then 1 g over 8 h
Single bolus (increasingly common in Canada)	2 g IV. In 2023, 60% of Ontario MHPs that included TXA used a 2 g dose (CJEM 2025)
Pediatric	15 mg/kg (max 1 g) bolus, then 2 mg/kg/h (per local pediatric protocol)

Benefit is time-dependent and absent after 3 hours. Ask EMS/STARS exactly what dose and when.

Laboratory targets during an MHP

Minimum targets from the Ontario provincial MHP toolkit ([ORBCoN toolkit](#)):

Parameter	Target	Treat with
Hemoglobin	> 80 g/L	Red cells
INR	< 1.8	Plasma (or PCC)
Fibrinogen	> 1.5 g/L (≥ 2.0 in obstetric hemorrhage)	Fibrinogen concentrate (4 g) or cryoprecipitate
Platelets	> $50 \times 10^9/L$ (> 100 in CNS injury)	Platelets
Ionized calcium	> 1.15 mmol/L	Calcium chloride or gluconate
Temperature	> 36 °C	Warmer, forced air, warm room

Fibrinogen is the most frequently forgotten. Give it early — levels fall first and fastest.

Calcium

Citrate in blood products binds calcium; trauma itself causes hypocalcemia. Low calcium worsens contractility, vascular tone and clotting.

- Empiric: **calcium chloride 1 g IV** (or calcium gluconate 3 g) with the first units, then guided by ionized calcium; many protocols repeat with every 2–4 units.
- Calcium chloride through a good line — it is a vesicant.

Permissive hypotension — targets

Situation	Target until hemorrhage control
Penetrating or blunt trauma, no TBI	SBP 80–90 mmHg (MAP 50–65); radial pulse and mentation
TBI	SBP ≥ 110 (age 15–49 or > 70); ≥ 100 (age 50–69)
Spinal cord injury	MAP ≥ 80 –85
Elderly, chronic hypertension	Individualize — they tolerate hypotension poorly

Permissive hypotension is a bridge to the OR, not a destination. The CBS guide notes that in hospital it is reasonable to target normal perfusion once hemorrhage control is under way.

Adjuncts and what's not (yet) routine

- **Vasopressors:** not a substitute for blood. A small dose may be needed post-intubation or in spinal shock.
- **PCC in non-anticoagulated trauma:** the PROCOAG RCT found no reduction in 24-h blood products and more thromboembolic events ([ORBCoN MHP 2.0 summary](#)). Reserve PCC for anticoagulant reversal or when plasma is unavailable.

- **Viscoelastic testing (ROTEM/TEG):** useful for guiding therapy after the first packs where available.
- **REBOA:** only in centres with a trained program; not a routine ED intervention in Alberta.

Script: the MHP running commentary

"Blood update please." → "Four red cells, four plasma, one platelets in, calcium one gram given, TXA at 14:40." → "Thank you. BP?" → "92 over 60." → "OK, responding. Next pack please. Let's get fibrinogen and ionized calcium back."

This 15-second exchange every 5 minutes is how you keep the ratio and the calcium honest.

Memory hook — Trigger · Team · TXA · Testing · Targets · Temperature · Terminate.

Pearl — Fibrinogen falls first and fastest. Order it with the first MHP bloods, not the third.

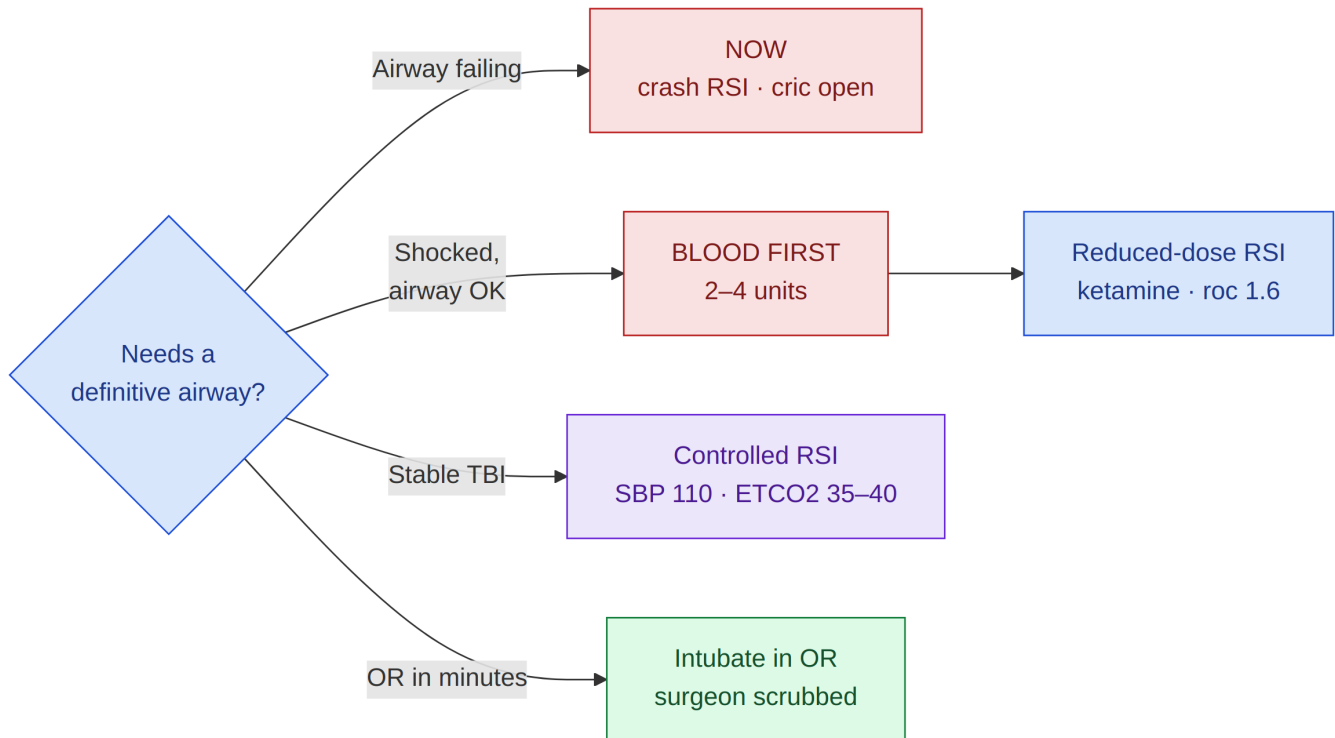
Pearl — Every minute of delay to the first cooler costs lives. Pre-alert blood bank before the patient arrives.

Pitfall — More than 1 L of crystalloid. Every bag of saline is a bag of coagulopathy.

Pitfall — Forgetting calcium. Hypotension that won't respond to blood may be hypocalcemia.

7. The airway in trauma

The trauma airway is usually physiologically difficult before it is anatomically difficult: hypovolemia, hypoxia and acidosis kill at induction. Decide **whether** and then **when** — and in the bleeding patient, when is usually "after the first units of blood."



Resuscitate before you intubate

Scenario	Timing
Airway obstructed, failing oxygenation despite adjuncts	Now — crash or modified RSI, cric ready
Shocked, airway maintained, sats OK	Blood first (2–4 units), then RSI with reduced doses
TBI with GCS ≤ 8 , stable	Controlled RSI with strict SBP and ETCO2 targets
Going to OR within minutes, airway maintained	Consider intubating in the OR with surgeon scrubbed
Combative, needs CT, cannot be safely sedated	RSI after primary survey and resuscitation

A shock index ≥ 0.9 predicts post-intubation hypotension and peri-intubation arrest. Say it out loud before induction.

Pre-intubation checklist (challenge and response)

"Airway checklist please — I'll read, you respond."



Pre-oxygenated (NRB + nasal cannula at 15 L/min, or BVM with PEEP); apneic oxygenation on



Two suctions working



Video laryngoscope and direct blade; bougie; tube sizes; SGA

- ☐ Cric kit open, neck landmarks marked
- ☐ Manual in-line stabilization assigned; collar front opened
- ☐ Two working IVs; blood running; push-dose pressor drawn
- ☐ Drugs and doses read out loud
- ☐ ETCO2 attached
- ☐ Plan A, B, C, D stated
- ☐ Post-intubation sedation and analgesia ready

Drug choices in the shocked trauma patient

Drug	Usual dose	In shock
Ketamine	1–2 mg/kg IV	0.5–1 mg/kg (or less); first choice for most
Etomidate	0.3 mg/kg	0.1–0.15 mg/kg
Fentanyl	1–3 mcg/kg	Omit or minimize
Propofol	1–2 mg/kg	Avoid
Rocuronium	1.2 mg/kg	1.6 mg/kg (slower circulation delays onset)
Succinylcholine	1.5 mg/kg	2 mg/kg; avoid if hyperkalemia risk (crush > 24 h, old burns, spinal cord injury > 72 h)
Push-dose epinephrine	10–20 mcg IV (10 mcg/mL)	Draw up before induction
Phenylephrine	100 mcg IV	Alternative

Doses are a guide; confirm with local pharmacy and your own practice.

Plan A → D (say it before you start)

"Plan A: video laryngoscope with bougie, two attempts maximum. Plan B: supraglottic airway and oxygenate. Plan C: bag-mask with two people and adjuncts. Plan D: surgical cric — scalpel, finger, bougie, size 6 tube. If sats drop below 90 at any point, I'll call it and we stop and oxygenate."

After the tube

- ETCO2 waveform confirmation, then chest X-ray for depth.
- Ventilation in hypovolemic shock: low PEEP (5), modest rate, tidal volume 6–8 mL/kg ideal body weight. High intrathoracic pressure kills preload.
- TBI: target ETCO2 35–40 mmHg (PaCO2 35–45); avoid hyperventilation unless herniating.
- Watch for new tension pneumothorax after positive pressure starts — re-scan for lung sliding if BP drops.
- Sedation and analgesia early (ketamine or fentanyl infusion as BP tolerates). Awareness under paralysis is common in trauma.

Special airway situations

- **Facial fractures / massive oral bleeding:** two suctions, patient upright if spine allows, "double setup" with neck prepped for cric.
- **Penetrating neck with expanding hematoma or voice change:** early intubation by the most experienced operator, awake approach if feasible, surgical team in the room.
- **Inhalation burns:** singed hair alone is not an indication; stridor, voice change, soot in the airway with respiratory distress, full-thickness face burns, or large TBSA burns are. Use a large tube (≥ 7.5 – 8.0).
- **C-spine:** manual in-line stabilization is enough; don't let the collar make laryngoscopy harder.

Memory hook — Blood before tube. Unless the tube *is* the emergency.

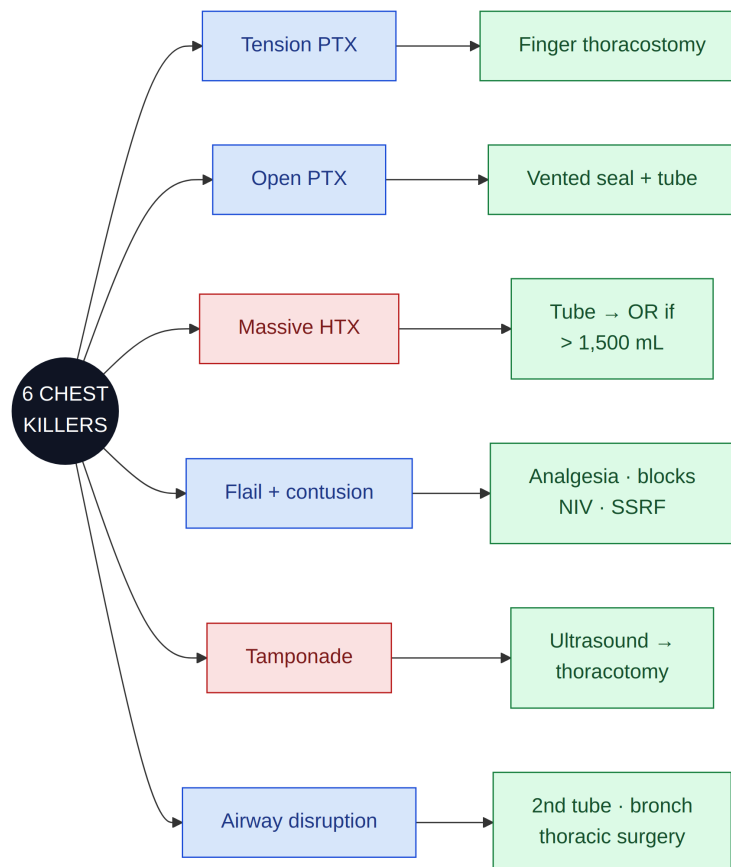
Pearl — Shock index ≥ 0.9 ? Say it out loud, draw up push-dose epinephrine, and halve the induction dose.

Pitfall — Positive pressure unmasks a tension pneumothorax. BP drops after the tube → re-scan both chests.

Pitfall — Paralysed but not sedated. Start post-intubation analgesia and sedation within minutes.

8. Breathing: the chest in the first hour

Six chest injuries kill in the primary survey: tension pneumothorax, open pneumothorax, massive hemothorax, flail chest with contusion, cardiac tamponade and airway disruption. The first three are fixed with a finger, a seal and a tube.



Tension pneumothorax

A clinical diagnosis. Hypotension + hypoxia + absent sliding on one side = decompress now. Don't wait for an X-ray.

Setting	Technique
Spontaneously breathing, stable enough	Needle decompression 4th/5th ICS just anterior to the mid-axillary line (8 cm catheter in adults); 2nd ICS mid-clavicular still acceptable, especially in children
Ventilated patient	Finger thoracostomy: 3–4 cm incision, 4th/5th ICS anterior to mid-axillary line, blunt dissect over the rib, finger sweep. Faster and more reliable than a needle
After decompression	Chest tube through the same hole

"[Name], left finger thoracostomy please. Tell me what you feel — air, blood, lung up?" ... "Rush of air, lung up." ... "Good. BP? Let's put a tube through that."

Open pneumothorax

Vented chest seal (or a dressing taped on three sides), then a chest tube away from the wound. Intubate if the patient is tiring.

Hemothorax and massive hemothorax

- Chest tube: 28–32 Fr is adequate for hemothorax (ATLS 10 reduced sizes); small-bore catheters are acceptable for simple pneumothorax.
- Consider autotransfusion if your department has the collection system and protocol.
- **Indications for urgent thoracotomy:** initial drainage > 1,500 mL, ongoing > 200 mL/h for 2–4 h, or persistent instability despite transfusion.

"Chest tube's in — how much blood?" ... "About 1,600." ... "That's a massive hemothorax. Trauma surgery, this patient needs the OR for a thoracotomy. Leave the tube unclamped and keep counting output. Next pack please."

Flail chest and pulmonary contusion

- Analgesia is the treatment: multimodal plus early regional (serratus anterior or erector spinae plane block) if your team can do them.
- Oxygenation support; NIV or HFNC in the awake cooperative patient without other contraindications.
- Avoid excess fluid (contused lung leaks).
- Discuss surgical rib fixation (SSRF) with trauma surgery for flail segments.

Blunt aortic injury (don't miss it in the first hour)

- Suspect with high-energy deceleration, widened mediastinum, apical cap, left hemothorax, loss of aortic knob, depressed left mainstem bronchus.
- Diagnosis: CT angiogram of the chest.
- Once other bleeding is controlled: **impulse control** — HR < 100 and SBP < 100–120 with a short-acting beta-blocker (esmolol) first, then vasodilator if needed. Coordinate with vascular/trauma surgery.

Blunt cardiac injury

ECG on everyone with significant chest trauma; troponin. Normal ECG and troponin essentially exclude clinically significant BCI. New arrhythmia or unexplained hypotension → echo.

Tracheobronchial injury

Suspect with a persistent large air leak after chest tube, pneumothorax that won't re-expand, subcutaneous emphysema, hemoptysis. May need a second tube, bronchoscopy, and single-lung ventilation. Call thoracic surgery early.

Memory hook — A finger, a seal and a tube fix three of the six.

Pearl — In a ventilated patient, a finger beats a needle: faster, more reliable, and you can feel the lung.

Pitfall — Waiting for a chest X-ray to confirm tension. It's a clinical diagnosis.

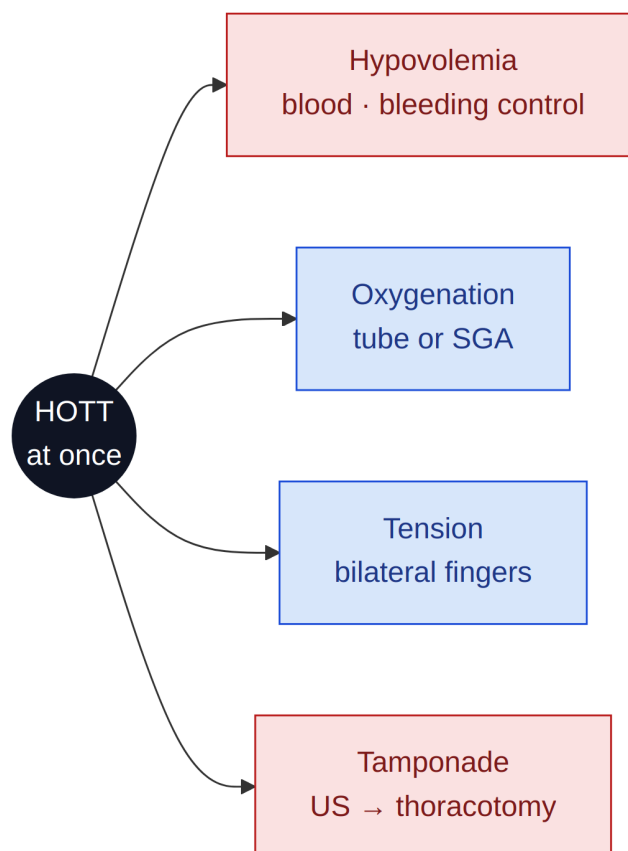
Pitfall — Elderly patients with "just rib fractures" — they deteriorate on day two. Treat the pain early.

9. Traumatic cardiac arrest and resuscitative thoracotomy

Traumatic arrest is not medical arrest with blood on the floor. Chest compressions do little for an empty or tamponaded heart; **reversing the cause comes before, or alongside, CPR**. This is where your ACLS instincts need to be overridden.

Your ACLS brain vs your trauma brain:

	Medical arrest	Traumatic arrest
First priority	Compressions and defibrillation	Reverse HOTT, all at once
Fluids	Minimal	Blood, fast, via rapid infuser
Chest	Leave it alone	Bilateral finger thoracostomies
Heart ultrasound	Pulse checks, cardiac activity	Tamponade? → open the chest
Epinephrine	Core drug	Low priority; may harm the empty heart
Where survival happens	In the ED	In the OR

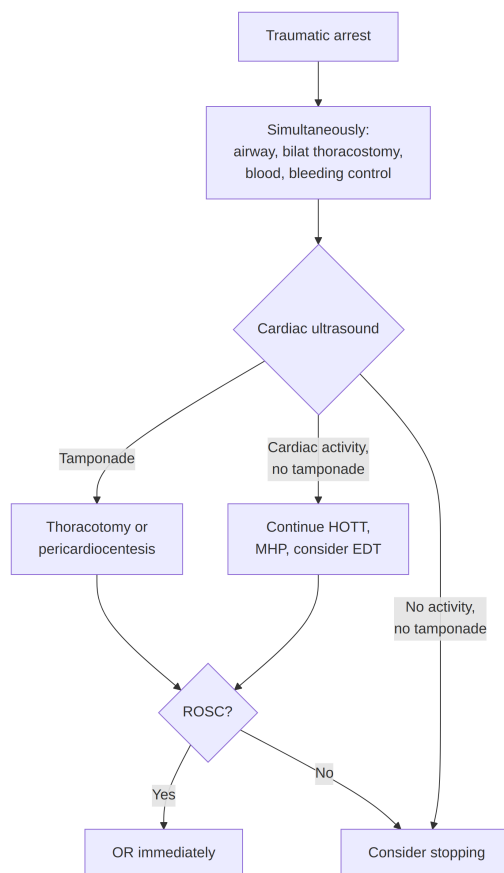


The four reversible causes: HOTT

Cause	Action (all simultaneously)
Hypovolemia	Control external bleeding, pelvic binder, blood via rapid infuser (MHP), TXA
Oxygenation	Airway and ventilation — intubate or SGA; don't delay other steps
Tension pneumothorax	Bilateral finger thoracostomies — don't wait for signs
Tamponade	Ultrasound the heart; pericardiocentesis as a bridge, thoracotomy definitively

Also think: was this a *medical* arrest that caused the crash (MI, arrhythmia, seizure)? If the mechanism is minor and the story doesn't fit, run a medical code too.

Algorithm



CPR and epinephrine are given per your team's practice but take lower priority than the HOTT interventions. Don't let compressions stop a thoracostomy or line.

Emergency department thoracotomy: who benefits?

EAST 2015 recommendations ([Seamon et al., J Trauma Acute Care Surg 2015](#)) for patients arriving **pulseless**:

Mechanism	Signs of life present	Signs of life absent
Penetrating thoracic	Strongly recommended	Conditionally recommended
Penetrating extrathoracic	Conditionally recommended	Conditionally recommended
Blunt	Conditionally recommended	Conditionally recommended against

Signs of life = pupillary response, spontaneous ventilation, carotid pulse, measurable BP, extremity movement, or cardiac electrical activity. Survival is best for isolated stab wounds to the heart. Western Trauma Association guidance also uses CPR duration: generally no EDT after > 10 minutes of prehospital CPR in blunt trauma or > 15 minutes in penetrating trauma without response.

Alberta reality: whether ED thoracotomy happens at RAH depends on who is in the room and the local trauma program's policy. Know your program's position before the day you need it, and have the tray location memorized.

Script: running a traumatic arrest

*"This is a traumatic arrest. We're going to treat reversible causes first.
[Name], airway — tube or SGA, whatever's fastest. [Name] and [Name], bilateral finger thoracostomies now. [Name], blood through the rapid infuser, MHP is activated. [Name], ultrasound the heart — tell me about tamponade and activity. Compressions continue unless they get in the way of a procedure."*

Then:

"Thoracostomies done — right had air, left had blood. Cardiac ultrasound shows [no tamponade / organized activity]. Two units in. Let's check a pulse."

When to stop

Consider stopping when all four HOTT causes have been addressed and there is no cardiac activity on ultrasound and no tamponade, or when time-based EDT thresholds have passed without response. State it clearly and invite objections:

"We've addressed hypovolemia, oxygenation, both chests and tamponade. There's no cardiac activity on ultrasound. I'm proposing we stop. Does anyone have a reason to continue?"

Memory hook — HOTT before CPR: Hypovolemia, Oxygenation, Tension, Tamponade.

Pearl — The best ED thoracotomy survivor is the young patient with a single stab wound to the heart and signs of life.

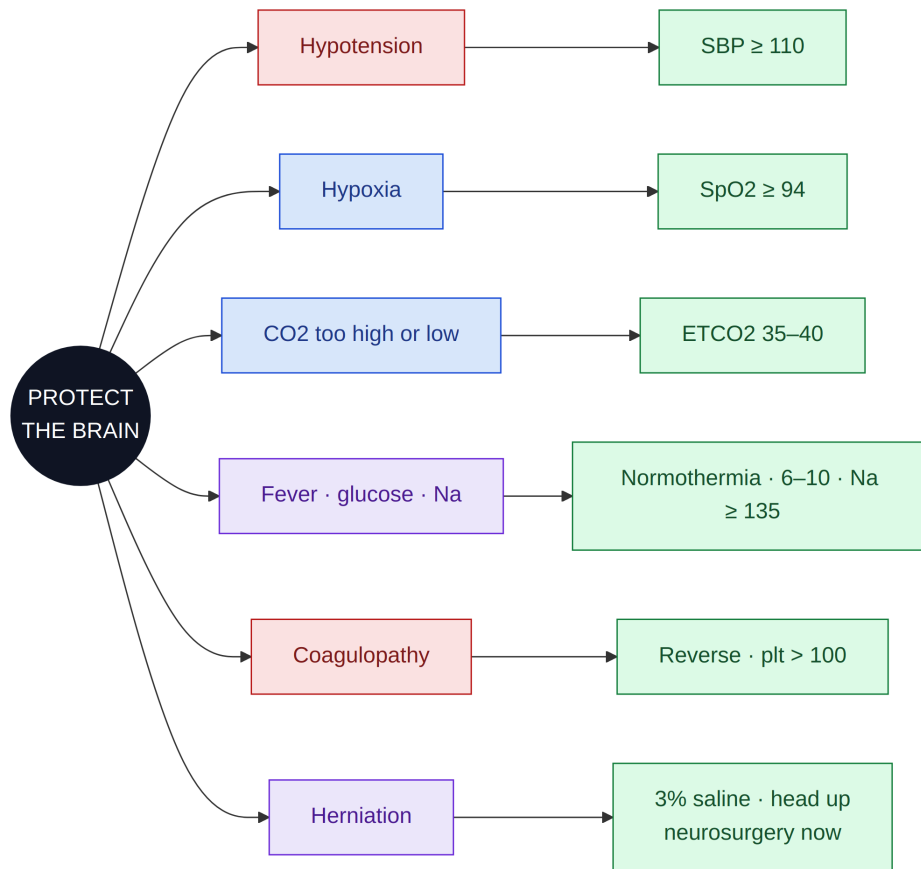
Pitfall — Running pure ACLS on a trauma patient. Compressions on an empty heart move nothing.

Pitfall — Not thinking "medical first": a minor crash with an arrest may be an MI that caused the crash.

10. Disability: head and spinal cord injury in the first hour

The first hour in TBI is about preventing secondary brain injury. A single episode of hypotension or hypoxia roughly doubles mortality — so the ED targets matter more than any single drug.

The brain's enemies — and the number that beats each one:



Neuroprotection targets ("the numbers")

Parameter	Target
SBP	≥ 110 mmHg (age 15–49 or > 70); ≥ 100 mmHg (age 50–69) — Brain Trauma Foundation
SpO2	≥ 94%; never < 90%
PaCO2 / ETCO2	PaCO2 35–45 mmHg; ETCO2 35–40 (lower only for active herniation)
Temperature	Normothermia; treat fever
Glucose	6–10 mmol/L
Sodium	≥ 135 mmol/L; no hypotonic fluids
Head position	Head of bed 30° (reverse Trendelenburg if spine not cleared); collar loosened; tube ties not compressing jugulars
Platelets / INR	Platelets > 100; INR corrected; reverse anticoagulants
Hemoglobin	Avoid anemia; > 70–90 g/L depending on context

"Everyone, this is a severe head injury. Our targets are SBP above 110, sats above 94, ETCO2 35 to 40. Recorder, call out any SBP under 110 or sats under 94 immediately."

Signs of herniation

Unilateral fixed dilated pupil, extensor posturing, decreasing GCS by ≥ 2 points, Cushing's response (hypertension, bradycardia, irregular breathing).

Bridge to neurosurgery:

1. **Hyperosmolar therapy** — 3% hypertonic saline 250–500 mL IV (or 23.4% saline 30 mL via central line), or mannitol 0.5–1 g/kg if not hypotensive.
2. **Brief hyperventilation** — ETCO₂ 30–35 as a temporary measure only, while the OR is being arranged.
3. **Deep sedation and analgesia**, head up, normothermia.
4. **Neurosurgery now** — and CT as soon as the patient is safe to move.

"Left pupil is blown. This is herniation. Hypertonic saline 250 now, ETCO₂ to 30–35 temporarily. Neurosurgery please — tell them we're heading to CT in 5 minutes."

Other TBI essentials

- **TXA:** CRASH-3 showed benefit for mild-moderate TBI when given within 3 hours; include TXA in the bleeding polytrauma patient with TBI.
- **Seizure prophylaxis:** discuss with neurosurgery for severe TBI (levetiracetam or phenytoin for 7 days is common practice).
- **Anticoagulated head injury:** reverse immediately — don't wait for CT if the patient is deteriorating (Section 11).
- **Where's neurosurgery?** Know whether cranial neurosurgery is on site on the day, or whether the patient needs to move to another centre via RAAPID. Decide this early.

Spinal cord injury

- **Suspect:** flaccid areflexia, priapism, diaphragmatic breathing, sensory level, hypotension with bradycardia and warm peripheries.
- **Neurogenic shock is a diagnosis of exclusion.** Rule out hemorrhage first — most hypotensive trauma patients with spinal injury are also bleeding.
- **Targets:** MAP ≥ 80 –85 mmHg. Norepinephrine is a reasonable first vasopressor; atropine or pacing for symptomatic bradycardia.
- **High cord injuries (C3–C5):** expect respiratory failure; intubate early and calmly rather than late and in crisis.
- **No steroids.**
- **Spine precautions:** limit motion, but don't let a collar compromise the airway, raise ICP, or cause pressure injury. Remove from the spine board promptly.

GCS reporting

Always report components, especially the best motor score, and note confounders (intoxication, sedation, intubation, hypotension, hypoxia). Record the pre-intubation GCS — neurosurgery will ask.

Memory hook — "110 · 94 · 35–40" — say it like a phone number.

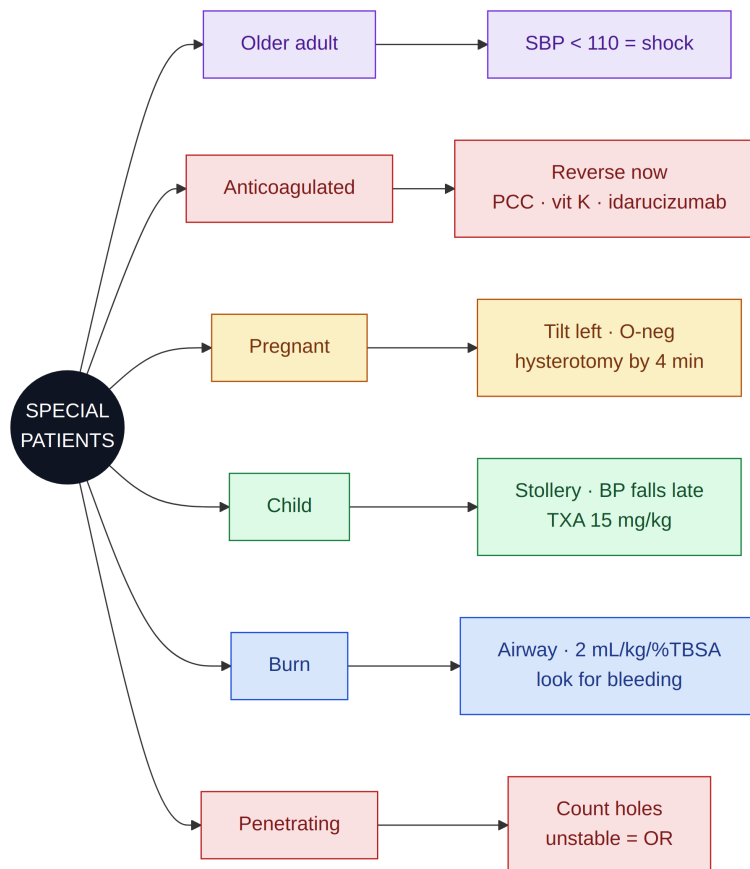
Pearl — One episode of hypotension or hypoxia roughly doubles TBI mortality. Your recorder should call out every breach.

Pitfall — Permissive hypotension in a head-injured patient. TBI overrides the bleeding target.

Pitfall — Calling it neurogenic shock before excluding hemorrhage.

11. Special populations

The primary survey doesn't change; the physiology, thresholds and destinations do. Say the special population out loud in your pre-brief so the team adjusts with you.



Older adults and anticoagulated patients

Older patients look well until they don't. "Normal" vitals may be shock.

- **Shock threshold:** treat SBP < 110 mmHg as hypotension in patients ≥ 65 . Beta-blockers and pacemakers blunt tachycardia. Use lactate and base deficit early.
- **Low-energy mechanisms matter:** a fall from standing on an anticoagulant is a trauma activation-level event for the head.
- **Rib fractures are lethal** in the elderly: each additional fracture raises pneumonia and mortality risk. Early analgesia, regional blocks, incentive spirometry, admission.
- **Lower threshold for CT** (head, C-spine, chest/abdomen/pelvis).
- **Goals of care:** ask early, ideally with family, before the patient reaches the ICU.

Anticoagulant reversal in major trauma or head injury (confirm with your pharmacy and Thrombosis Canada guidance):

Agent	Reversal
Warfarin	4-factor PCC (Octaplex or Beriplex) — fixed doses of about 1,500–2,000 IU are increasingly common, or weight/INR-based per local protocol — plus vitamin K 10 mg IV
Apixaban, rivaroxaban, edoxaban	4-factor PCC, commonly 2,000 IU (about 25–50 IU/kg) per local protocol; andexanet availability varies
Dabigatran	Idarucizumab 5 g IV
Heparin	Protamine
Antiplatelets	Platelet transfusion is not routinely recommended in ICH unless neurosurgery is planned — discuss with neurosurgery

Weight-based fixed dosing in Canada has historically used bands such as 1,000 IU (< 40 kg), 1,500 IU (41–60 kg), 2,000 IU (61–80 kg), 2,500 IU (81–100 kg) and 3,000 IU (> 100 kg) ([Canadian cohort](#)).

"He's on apixaban with a head bleed. Let's give PCC 2,000 units now — don't wait for the CT read if he's deteriorating. [Name], please call pharmacy."

Pregnancy

Two patients; the best fetal resuscitation is maternal resuscitation.

- **Physiology:** HR up 10–15, BP lower in the second trimester, blood volume up ~40%. Mothers may lose 30–35% of blood volume before becoming hypotensive. Normal PaCO₂ is about 30, so 40 means respiratory failure.
- **Position:** from ~20 weeks (fundus at the umbilicus), manual left uterine displacement or a 15–30° left tilt on the spine board.
- **Blood:** O-negative until group is known.
- **Imaging:** don't withhold CT that the mother needs.
- **Rh-negative mothers:** Rh immunoglobulin; Kleihauer-Betke test.
- **Fetal monitoring** from ~20–24 weeks for at least 4–6 hours after significant trauma; abruption can be delayed.
- **Call obstetrics and neonatology early.**
- **Maternal arrest ≥ 20 weeks:** resuscitative hysterotomy (perimortem caesarean) begins by 4 minutes of arrest, aiming for delivery by 5 minutes, in the resuscitation bay.

"She's about 28 weeks by fundal height. Manual uterine displacement please. OB and NICU to the trauma bay. If she arrests, we go to hysterotomy at 4 minutes — someone find the tray now."

Children

- **Destination:** major pediatric trauma in the Edmonton zone goes to the Stollery. Stabilize and transfer via RAAPID if a child lands at RAH or NECHC.
- **Weight:** length-based tape, or estimate (age + 4) × 2 kg for ages 1–10.
- **Shock:** tachycardia and delayed capillary refill come first; hypotension is a late, pre-arrest sign.
- **Volume:** limit crystalloid; switch to blood early (10–20 mL/kg per product, ratio-based pediatric MHP).
- **TXA:** 15 mg/kg (max 1 g), then 2 mg/kg/h.
- **Airway:** cuffed tubes; large occiput (pad under shoulders); cric is avoided in young children (needle cric < ~10–12 years).
- **Needle decompression:** 2nd ICS mid-clavicular line remains reasonable in children.

- **Non-accidental injury:** consider it with any mismatch between story and injury, especially under 2 years.
- **Course:** [TRIK — Trauma Resuscitation in Kids](#).

Burns

- **Airway first** (Section 7). Think carbon monoxide and cyanide in enclosed-space fires: 100% oxygen; hydroxocobalamin 5 g IV for suspected cyanide with shock or lactate > 8.
- **Fluids (ATLS 10/11):** 2 mL/kg/%TBSA in adults (3 mL/kg in children; 4 mL/kg for high-voltage electrical), half in the first 8 hours from time of burn. Titrate to urine output 0.5 mL/kg/h in adults (1 mL/kg/h in children < 30 kg).
- **Trauma first:** hypotension in the first hour after a burn is not burn shock — look for bleeding.
- **Escharotomy** for circumferential chest or limb burns compromising ventilation or perfusion.
- **Keep warm**, cover with clean dry dressings, early burn centre consultation via RAAPID.

Penetrating trauma

- **Expose everything:** axillae, perineum, scalp, back, gluteal folds. Count the holes (an odd number means a retained projectile or a missed wound).
- **Unstable with a torso wound = OR**, not CT.
- **Don't remove impaled objects** in the ED.
- **Neck:** hard signs (expanding or pulsatile hematoma, active bleeding, bruit, air bubbling, stridor, shock) → OR. Otherwise, CT angiography.
- **Evidence:** cut around bullet or stab holes in clothing, bag clothing separately, document wounds by description (not "entry"/"exit"). Police will ask.

Memory hook — Old hides it, kids crash late, mums compensate, burns distract, bullets wander.

Pearl — In pregnancy, the best fetal resuscitation is maternal resuscitation.

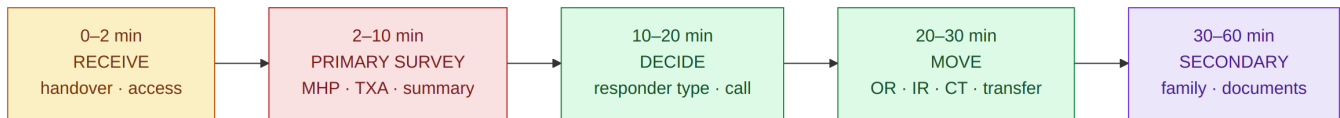
Pitfall — Waiting for the CT read before reversing an anticoagulated, deteriorating head injury.

Pitfall — Blaming early hypotension on the burn. Burn shock takes hours; bleeding takes minutes.

12. Decision points: minutes 10 to 60

The first hour ends with a destination. By 15–20 minutes you should be able to say where the patient is going and why. If you can't, that is the problem to solve next.

The clock on the wall:



The first-hour timeline

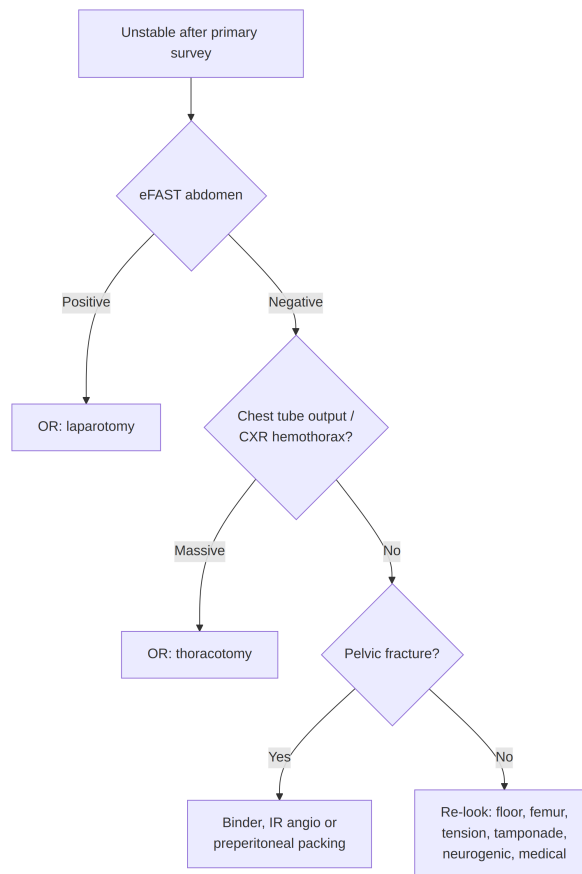
Time from arrival	Milestones
0–2 min	Hands-off handover, transfer, monitors, access
2–10 min	xABCDE complete, eFAST, MHP activated if needed, TXA given, primary survey summary spoken
10–20 min	Response to resuscitation assessed; destination decided; consultants called; transfer requested if needed
20–30 min	Patient leaves for OR/IR/CT, or is resuscitating toward it
30–60 min	Secondary survey (if stable), tertiary plans, family, documentation

An unstable patient still in your resuscitation bay at 30 minutes needs a spoken reason.

Response to resuscitation

Response	Pattern	Usual destination
Rapid responder	Normalizes after initial blood, stays normal	CT, then definitive care
Transient responder	Improves, then deteriorates again	OR or IR; CT only if the surgeon agrees and it's close by
Non-responder	No improvement despite blood	OR now (or IR for pelvic bleeding); if at a non-surgical site, transfer immediately

Where is the bleeding and where does the patient go?



Stable patients go to CT (pan-scan if the mechanism warrants it). The resuscitation continues in CT: a nurse and physician go with the patient, blood hangs, the monitor is watched.

At a non-surgical site (for example NECHC)

- The decision to transfer should be made in the first 15 minutes, based on physiology and mechanism.
- **Don't delay transfer for CT** that won't change your management. The receiving trauma centre will scan.
- Call RAAPID North early: "Major trauma for the trauma centre." Say your ETA to readiness, products given and airway status.
- Send blood with the patient if possible, along with a copy of all imaging and a short written summary including times of TXA, blood units and tourniquets.

The 5-minute reassessment script

Say a short version every 5 minutes, and a full version whenever something changes.

"Quick update, everyone. Time since injury is 50 minutes. BP now 95 over 60 after four packs. FAST positive in the right upper quadrant. TXA and calcium given, fibrinogen pending. He's a transient responder. Trauma surgery is taking him to the OR. [Name], please call OR to say we're 5 minutes away. [Name], pack the next MHP cooler to come with us. Anything I've missed?"

The call to the trauma surgeon (SBAR)

"Hi, it's AJ in the RAH resus bay. I have a 34-year-old male, motorcycle versus car at about 60 km/h, injured 40 minutes ago.

S: He's in hemorrhagic shock — BP 80 over 50, HR 135, MHP activated, four units in, transient response. **B:** GCS 14, FAST positive in Morison's pouch, pelvis X-ray normal, chest clear. **A:** I think he's bleeding from his liver or spleen and needs a laparotomy. **R:** I'd like you to come now and take him to the OR. Can you confirm you're coming, and should I call the OR?"

The recommendation line is what most calls lack. Name exactly what you want and by when.

Handover to the OR, ICU or receiving team

Use the same ATMIST structure you asked EMS to give you, plus: products given with times, TXA time, calcium and last labs, tourniquet times, airway grade and tube size, drugs and doses given, what hasn't been done (secondary survey, tertiary imaging), and family status.

Memory hook — Receive · Survey · Decide · Move. Two, ten, twenty, thirty.

Pearl — The "R" in SBAR is where calls succeed or fail. Say exactly what you want and by when.

Pitfall — CT is where unstable patients go to die. A transient responder goes to CT only if the surgeon agrees.

Pitfall — At a community site, scanning before transfer. The trauma centre will scan again anyway.

13. Imaging rules, secondary and tertiary survey

Once the patient is stable, two Canadian decision rules (both by Stiell and colleagues, derived and validated in Canadian EDs) cover most of your imaging choices for the minor and moderate trauma that fills the rest of the shift.

Canadian CT Head Rule

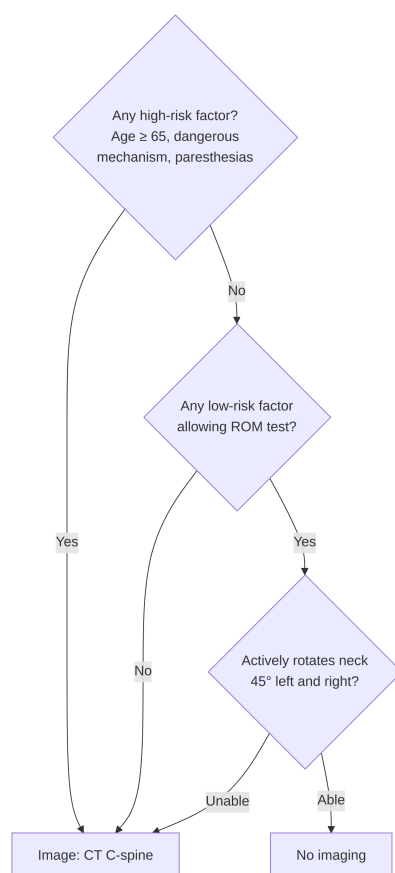
Applies to: GCS 13–15 after head injury with witnessed loss of consciousness, definite amnesia or witnessed disorientation. **Not for:** age < 16, anticoagulants or bleeding disorder, post-traumatic seizure, obvious open skull fracture, GCS < 13, unstable vitals.

CT head is required if **any** of:

Risk	Criteria
High (neurosurgical intervention)	GCS < 15 at 2 h after injury; suspected open or depressed skull fracture; any sign of basal skull fracture (hemotympanum, raccoon eyes, Battle's sign, CSF oto/rhinorrhea); vomiting ≥ 2 episodes; age ≥ 65
Medium (brain injury on CT)	Amnesia for events ≥ 30 min before impact; dangerous mechanism (pedestrian struck, ejection, fall > 3 ft or > 5 stairs)

Canadian C-Spine Rule

Applies to: alert (GCS 15), stable trauma patients.



- **Dangerous mechanism:** fall ≥ 3 ft or 5 stairs, axial load to the head, high-speed MVC (> 100 km/h), rollover or ejection, motorized recreational vehicle, bicycle collision.
- **Low-risk factors:** simple rear-end MVC, sitting in the ED, ambulatory at any time, delayed onset of neck pain, no midline C-spine tenderness.

Other imaging in the first hour

- **Pan-scan CT** (head, C-spine, chest/abdomen/pelvis with contrast) for high-energy mechanisms in stable patients.
- **CT angiogram of the neck** for seatbelt sign on the neck, cervical fractures involving the foramen transversarium, Horner's, unexplained neuro deficit (blunt cerebrovascular injury).
- **Pregnancy:** don't withhold needed imaging.

Secondary survey (when the patient is stable)

History: **AMPLE** — Allergies, Medications (especially anticoagulants and antiplatelets), Past history and pregnancy, Last meal and tetanus, Events and environment.

Head-to-toe: scalp, face, eyes (acuity, pupils, globe injury), ears (hemotympanum), nose, mouth and teeth; neck; chest; abdomen; pelvis and perineum; rectal/vaginal only if indicated; back during log-roll; all extremities (pulses, compartments, open fractures, joint stability); full neuro exam.

Don't forget: tetanus, antibiotics for open fractures (cefazolin within 1 hour; add gram-negative cover for severe contamination per local protocol), reduce and splint deformities with neurovascular checks before and after, analgesia.

"We're stable enough for the secondary survey. [Name], head to toe please, and call out every finding for the recorder. Let's log-roll at the back exam — I'll call it."

Tertiary survey

Repeat a full head-to-toe exam within 24 hours (and when the patient is awake). Missed injuries in trauma are common — especially hand, foot, spine and small-joint fractures. Document in the plan: "tertiary survey required."

Memory hook (CT Head high risk) — "G-O-B-V-O": GCS < 15 at 2 h · Open or depressed skull · Basal skull signs · Vomiting × 2 · Old (≥ 65).

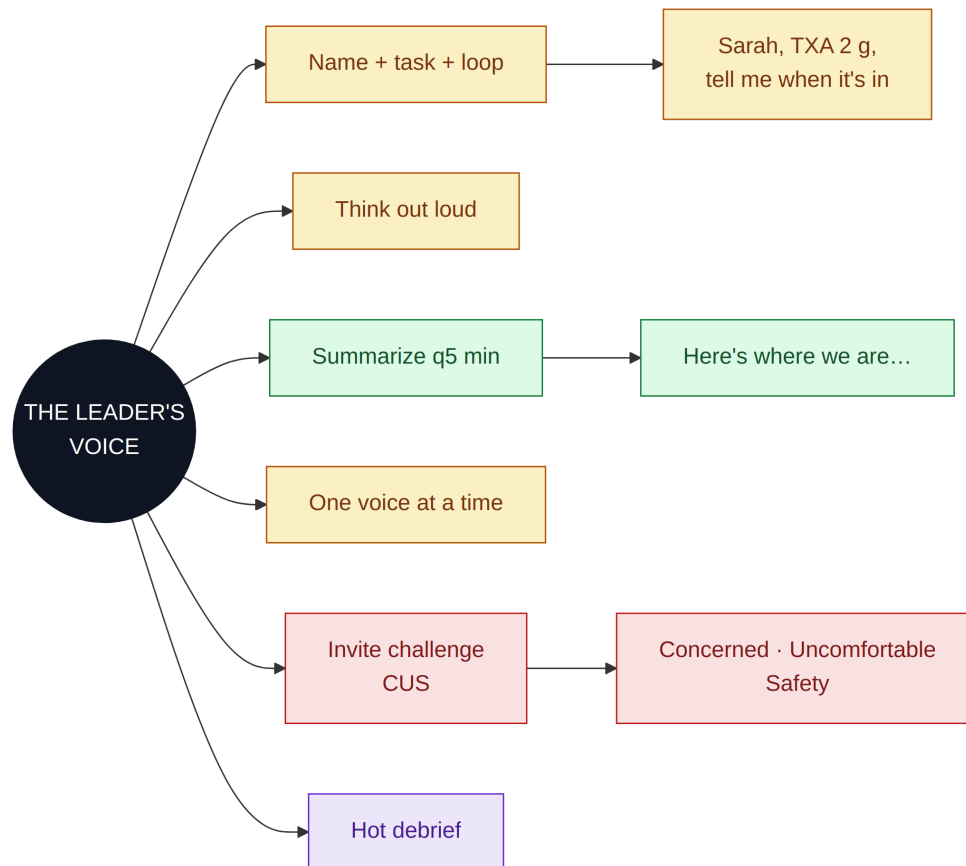
Memory hook (C-Spine) — High risk? Image. Low risk? Test. Turns 45° both ways? Clear.

Pitfall — Applying the CT Head Rule to anticoagulated patients. They are excluded — scan them.

Pearl — Write "tertiary survey required" in every trauma admission plan. Missed hand and foot fractures are the classic miss.

14. The team leader's phrase bank

You already have the words for medical resuscitation because ACLS gives you a rhythm. These phrases give trauma the same rhythm. Pick the ones that sound like you and practise them out loud until they're automatic.



Principles

- **Name + task + closed loop.** "Sarah, give TXA 2 grams IV, tell me when it's in." → "TXA in." → "Thank you."
- **Think out loud.** Your reasoning is the team's shared mental model.
- **Summarize on a schedule.** Every 5 minutes, and whenever something changes.
- **One voice at a time.** You control the volume of the room.
- **Invite challenge.** Psychological safety catches your errors.

Phrases by situation

Situation	Say
Starting	"I'm AJ, I'm leading. Everyone say your name and role please."
Getting attention	"Can I have eyes up for ten seconds."
Room too loud	"Let's keep it quiet — only the team leader and the person reporting."
Hands-off handover	"Hands off for thirty seconds — paramedics, go ahead."
Assigning	"[Name], can you [task], and tell me when it's done."
Checking progress	"[Name], where are we with the second line?"
Nobody responded	"I didn't hear a response — who has the TXA?"

Situation	Say
Starting the primary survey	"Airway, tell me about the airway."
Calling a diagnosis	"This is hemorrhagic shock until proven otherwise."
Activating MHP	"I'm activating the massive hemorrhage protocol. [Name], call blood bank. [Name], you own the blood."
Deterioration	"BP's dropped. Stop. Let's go back to the primary survey — airway, breathing..."
Deciding	"My plan is the OR. Does anyone disagree?"
Overwhelmed	"Everyone pause for ten seconds while I think." (the "10-for-10")
Need more help	"I need another physician in here now — can someone call [name]?"
Consultant arrives	"Welcome. Here's the summary: [30 seconds]. Would you like to take over leadership, or should I continue and you focus on the operative decision?"
Learner is leading, going off track	"Can I make a suggestion — let's check the chest again." Then, if needed: "I'm going to take over leadership now; thank you."
Disagreement	"I hear you. Tell me what you're seeing." Then decide and state it.
Doing a procedure yourself	"I'm going to do the chest tube. [Name], you're leading until I'm done."
Stopping the MHP	"Bleeding is controlled. I'm terminating the MHP — please tell blood bank and return unused products."
Ending	"Thank you, everyone. That was excellent work. Can we have a quick two-minute debrief once the patient leaves?"

Speaking up (for your team to use with you)

Tell your team at the pre-brief that you want to hear concerns. Teach the CUS escalation:

- **C** — "I'm **concerned** about ..."
- **U** — "I'm **uncomfortable** with ..."
- **S** — "This is a **safety** issue — we need to stop."

When someone uses it, stop and listen. Thank them even if you disagree.

What *not* to say

- "Someone get a line." (No name → nobody does it.)
- "Can we maybe think about blood?" (Hedging → no action.)
- Silence while you do a procedure. (The room loses its leader.)
- "Is he going to CT?" (A question when a decision is needed.)

The two-minute hot debrief

Right after the patient leaves:

1. "What went well?"
2. "What could we do better next time?"
3. "Any equipment or system problem we need to fix?"
4. "Is everyone OK?" — especially after a death, a child, or a violent injury.

Write system issues down and send them to your department's trauma or quality lead.

Practice drill — Pick five phrases from the table. Say each out loud three times in the resus bay on a quiet shift. Your mouth needs to have said them before your brain needs them.

Pearl — Silence is a signal. When the room goes quiet, the team is waiting for you to decide.

Pitfall — "Someone get a line." No name, no action.

15. Pocket card: the first hour on one page

Print this section or keep it on your phone. It's the whole playbook compressed.

Before arrival

1. Whiteboard: ATMIST, time of injury.
2. What kills first? Where do they go? Who do I need?
3. Pre-brief (60 s): roles, main worry, if-then plans (shock → MHP; airway → resus first; arrest → HOTT).
4. Room: warm, rapid infuser, O blood, TXA, calcium, binder, tourniquets, VL/bougie/cric, chest tray, US.

Arrival

1. "Hands off, thirty seconds." Ask: trajectory, worry, tourniquet time, blood/TXA given.
2. Slide on airway's count. Monitors, access, bloods, warm.

Primary survey — ask, hear, act, confirm

Letter	Key question	Key action
x	"Any major external bleeding?"	Tourniquet, pack, binder
A	"Tell me about the airway."	Suction, jaw thrust, plan the tube <i>after</i> blood
B	"Breath sounds and sliding?"	Finger thoracostomy for tension
C	"HR, BP, shock index, FAST?"	MHP, TXA 2 g, CaCl 1 g, warm
D	"GCS by component, pupils?"	SBP $\geq$ 110, SpO ₂ $\geq$ 94, ETCO ₂ 35–40
E	"Everything off, front and back."	Temperature, cover, warm

Key numbers

Item	Number
Shock index concerning	> 1
Permissive hypotension (no TBI)	SBP 80–90
TBI SBP	≥ 110 (≥ 100 age 50–69)
Spinal cord MAP	≥ 80 –85
Elderly hypotension	SBP < 110
MHP targets	Hb > 80 · INR < 1.8 · fibrinogen > 1.5 · plt > 50 (> 100 CNS) · iCa > 1.15 · T > 36
TXA window	< 3 h from injury
Massive hemothorax	$> 1,500$ mL initial or > 200 mL/h
RSI in shock	Ketamine 0.5–1 mg/kg · roc 1.6 mg/kg · push-dose pressor ready
Burn fluids (adult)	2 mL/kg/%TBSA, half in 8 h, UO 0.5 mL/kg/h
Pregnancy arrest	Hysterotomy by 4 min if ≥ 20 wk

Traumatic arrest — HOTT at once

Hypovolemia (blood, bleeding control) · Oxygenation (airway) · Tension (bilateral finger thoracostomies) · Tamponade (US → thoracotomy/pericardiocentesis).

Decide by 15-20 minutes

- Non-responder / transient + FAST positive → **OR**
- Massive hemothorax → **OR**
- Pelvic bleeding → **binder + IR/packing**
- Stable → **CT**
- Wrong hospital → **RAAPID North 1-800-282-9911**, don't delay for CT

Every 5 minutes, say:

"Time since injury _____. Vitals _____. What we've found _____. What we've given _____. Leading concern _____. Plan _____. Next three tasks _____. Anything I've missed?"

Self-test: cover the page and answer out loud

- ☐ Draw the master brain map from memory, with colours.
- ☐ Say the full pre-brief for a hypotensive motorcycle crash, ETA 8 minutes.
- ☐ Name the 7 Ts and one action for each.
- ☐ Recite the MHP targets: Hb, INR, fibrinogen, platelets, iCa, temperature.
- ☐ Give the RSI doses for a shocked 80 kg patient.
- ☐ List the six chest killers and the fix for each.
- ☐ Run HOTT for a penetrating arrest — who does what?
- ☐ Say "110 · 94 · 35-40" and what each number is.
- ☐ Give the end-of-primary-survey summary for a case from your last shift.
- ☐ Make the SBAR call to trauma surgery, ending with a clear recommendation.

Study tip — Tick a box only when you can do it without looking. Untick them all each month and start again.

16. Sources and further resources

Pages opened for this playbook are linked below. Doses and thresholds not linked to a source are standard teaching (ATLS, Brain Trauma Foundation, Canadian decision rules) and should be checked against current AHS protocols, RAH trauma program policies and pharmacy.

Alberta

- [Trauma Care Alberta — AHS / Acute Care Alberta](#) — provincial program, 10 trauma centres, registry criteria, education
- [RAAPID — AHS](#) — transfer and referral numbers
- [RAH Alberta Trauma Services — AHS](#)
- [CBC: Edmonton's two trauma centres \(Jan 2025\)](#)
- [STARS history and Blood on Board · STARS MHP \(2023\) · STARS TOWAR whole blood \(2025\)](#)
- [Edmonton 2010 Trauma Report](#)

Canadian transfusion and MHP

- [Canadian Blood Services Clinical Guide to Transfusion, Ch. 11: Massive hemorrhage](#)
- [CBS FAQ: Whole blood, leukocytes reduced \(2025\)](#)
- [Ontario Provincial Massive Hemorrhage Toolkit \(ORBCoN, 2021\)](#)
- [Ontario MHP 2.0: major changes \(ORBCoN, 2025\)](#)
- [MHP adoption follow-up survey, CJEM 2025](#)
- [Callum JL et al. A regional massive hemorrhage protocol developed through a modified Delphi technique. CMAJ Open 2019;7:E546-61](#)
- [Treat the Bleed](#)
- [Canadian 4F-PCC fixed weight-based dosing cohort](#)

Guidelines and courses

- [ATLS 11 — American College of Surgeons · ACS Brief, Sept 2025](#)
- [EAST ED thoracotomy guideline \(Seamon et al., 2015\)](#)
- [Trauma Association of Canada — ATLS course listings](#)
- [TRIK — Trauma Resuscitation in Kids](#)
- [Brain Trauma Foundation Guidelines for the Management of Severe TBI, 4th edition](#)
- [Stiell IG et al. The Canadian CT Head Rule \(Lancet 2001\) and The Canadian C-Spine Rule \(JAMA 2001\)](#)

Canadian learning resources

- [Emergency Medicine Cases: Massive Hemorrhage Protocols 2.0 \(Helman, Callum, Petrosoniak, 2025\)](#)
- [Emergency Medicine Cases: The 7 Ts of Massive Hemorrhage Protocols \(2021\)](#)

Open questions to confirm locally

- ☐ AHS age cut-off for O-negative emergency red cells in females
- ☐ AHS/RAH MHP pack contents and TXA dose (1 g + 1 g vs 2 g bolus)
- ☐ RAH policy on ED resuscitative thoracotomy and REBOA

☐

Where cranial neurosurgery is on call on any given day, and the RAH → UAH pathway

☐

Pediatric age cut-off for Stollery vs adult centre

☐

Availability of viscoelastic testing and fibrinogen concentrate in the RAH ED